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Tunable multi-channel optical filter based on the near-infrared ultralow-loss phase change material Sb_2Se_3

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Abstract: Chalcogenide phase-change material ($Ge_2Sb_2Te_5$) has attracted considerable research interest in recent years for tunable structural color applications, owing to its pronounced contrast in optical properties between amorphous and crystalline states. However, the strong absorption of conventional $Ge_2Sb_2Te_5$ has significantly limited its applicability in the near-infrared (NIR) wavelength range. Herein, we utilize the near-zero absorption characteristic of the emerging phase-change material Sb_2Se_3 in the NIR band to construct a Fabry–Pérot cavity structure. This structure not only enables resonance peaks with high transmittance and large modulation range, but also allows a multi-channel filter to be achieved by increasing the thickness of Sb_2Se_3. This tunable multi-channel spectrally selective filter presents significant application potential in advanced fields such as multi-gas sensing, laser beam combining, and adaptive multi-spectral imaging.

1. Introduction

Fabry-Perot (F-P) cavities are a typical optical resonant structure [1]. By allowing the incident light to continuously reflect within the medium cavity between two optical mirrors, a multi-beam interference phenomenon is formed, thereby achieving the selection and enhancement of the specific wavelength light [2–4]. Due to its low process complexity and low loss characteristics, it is widely applied in various photonic devices such as structural color [5–7], lasers [8–10], and sensors [11,12]. However, the optical properties of the traditional F-P cavity remain unchanged after preparation, which significantly limits its application in tunable photonic systems. The chalcogenide phase-change material (PCM, such as $Ge_2Sb_2Te_5$) has become an ideal medium material for the tunable structure color of F-P cavities [13], due to the significant changes in optical constants [14], as well as its rapid [15,16], reversible, and high durability [17,18] characteristics during the phase transition process.

Since Harish [14] first used the PCM material in phase-change memory to construct tunable phase-change F-P cavities in 2014, researchers have conducted extensive exploration of its applications in smart window [19,20], micro display [21–23], and multi-spectral camera [24,25]. However, due to the high conductivity of the crystalline $Ge_2Sb_2Te_5$, which leads to the absorption of free carriers, there is significant optical loss in the near-infrared (NIR) wavelength [26,27]. This severely limits the application scope of the phase-change F-P cavities. However, the Sb_2S_3 and Sb_2Se_3 materials, which were once overlooked in the field of phase change materials due to their low conductivity, due to the fact that near-zero loss in the NIR band has gradually become the research hotspot for various phase-change tunable photonic systems [28–33]. In 2020, Otto L designed Sb_2S_3 and Sb_2Se_3 phase-change optical waveguides, which exhibited a transmission loss at 1550 nm that was two orders of magnitude lower than that of $Ge_2Sb_2Te_5$ and more than

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4000 cycles [34]. In 2022, Carlos and many other researchers have also pointed out that the low loss characteristics of Sb_2S_3 and Sb_2Se_3 in the NIR band make them ideal materials for programmable photonic devices in the non-visible light band [35–38]. Although significant achievements have been made in the field of optical waveguides, their dynamic modulation capability within the F-P cavity remains critically underexplored. Moreover, a design for a tunable spectrally selective filter in the NIR that simultaneously achieves low loss and wide modulation range is still lacking.

In this paper, the low-loss phase-change material Sb_2Se_3 was used to construct F-P cavities. A tunable narrow-band filter with high transmittance, wide modulation range, and adjustable number of transmission peaks has been successfully designed in the NIR band. Sb_2Se_3 is used as the tunable transparent medium layer, due to its low-k and large Δn characteristics in the NIR wavelength range. The transmittance of the crystalline state sample exceeds 45%, and the peak position shifts by more than 150 nm. By increasing the thickness of the Sb_2Se_3 layer, the F-P cavity can generate multiple transmittance peaks and adjust the peak positions and number. This characteristic of multi-channel and narrow-band filtering makes it highly promising for applications in areas such as gas sensors [39,40], laser beam combining [41,42], and multi-spectral imaging [43,44].

2. Simulation and experiment

In the simulation part, the finite element simulation software was used to design and optimize the multi-layer film structure. In the model construction, the substrate is 1μm SiO_2. Among them, the optical parameters of SiO_2, Ag and In_2O_3 are derived from the software database. The optical parameters of amorphous and crystalline Sb_2Se_3 were obtained through ellipsometry (RC2 XI, J. A. WOOLLAM) measurements. In the film preparation part, multi-layer film samples were grown using a magnetron sputtering device (TRP-450, SKY Technology Development Co., Ltd.). The pressure during film deposition is 0.5 Pa, and the Ar gas flow rate is 60 sccm. Among them, Ag uses a direct current power supply, with a sputtering power of 25W. In_2O_3 uses a direct current power supply, with a sputtering power of 40W. Sb_2Se_3 is processed using a radio frequency power supply, with a sputtering power of 30W. $Ge_2Sb_2Te_5$ uses a direct current power supply, with a sputtering power of 30W.

The deposited state will be regarded as the amorphous sample, and the sample obtained after 310°C, 30 min annealing in a tube furnace will be regarded as the crystalline state. UV-Vis-NIR spectrophotometer (Lambda 1050+, PerkinElmer 1) is used to test the visible-NIR spectra of multi-layer film samples.

3. Results and discussion

The traditional phase-change material $Ge_2Sb_2Te_5$ exhibits strong absorption of NIR light due to its high carrier concentration in the crystalline state. Reducing the thickness of $Ge_2Sb_2Te_5$ is an effective way to increase the transmission rate in the NIR band, but this often leads to a significant decline in the spectral modulation capability. Compared with $Ge_2Sb_2Te_5$, Sb_2Se_3 has better adaptability to the spectral modulation in the NIR band.

As shown in Fig. 1(a), the optical constants of $Ge_2Sb_2Te_5$ and Sb_2Se_3 are presented in the wavelength range of 400–1600 nm. It can be observed that the k value of Sb_2Se_3 is close to 0 in the NIR wavelength range above 800 nm. This means that Sb_2Se_3 can serve as a NIR transparent active dielectric layer, enabling the design of high-performance tunable NIR filters. This paper designs a typical F-P cavity structure based on Sb_2Se_3. As shown in Fig. 1(b), using Ag as the semi-transparent and semi-reflective layer, Sb_2Se_3 as the intermediate dielectric layer. However, the refractive index of Ag ($0.1 < n < 0.5$), Sb_2Se_3 ($n > 3$), and air ($n = 1$) differ significantly. Such strong refractive index mismatch at interfaces can lead to significant Fresnel reflection losses, which are detrimental to efficient spectral performance. In contrast, the n of In_2O_3 ranges from

0.8 to 1.7, enabling it to act as an interface buffer layer that reduces interface reflection losses. Therefore, the In_2O_3 layer is introduced to improve impedance matching at both the Ag/air and Ag/Sb_2Se_3 interfaces. The transmission peak position is controlled by the refractive index change brought about by the phase transition of Sb_2Se_3. Based on the above design principles, after conducting extensive simulations and explorations on the film structure, a multilayer film structure of In_2O_3-Ag-In_2O_3-SbSe-In_2O_3-Ag-In_2O_3 was designed. The thickness of each layer is shown in Table 1.

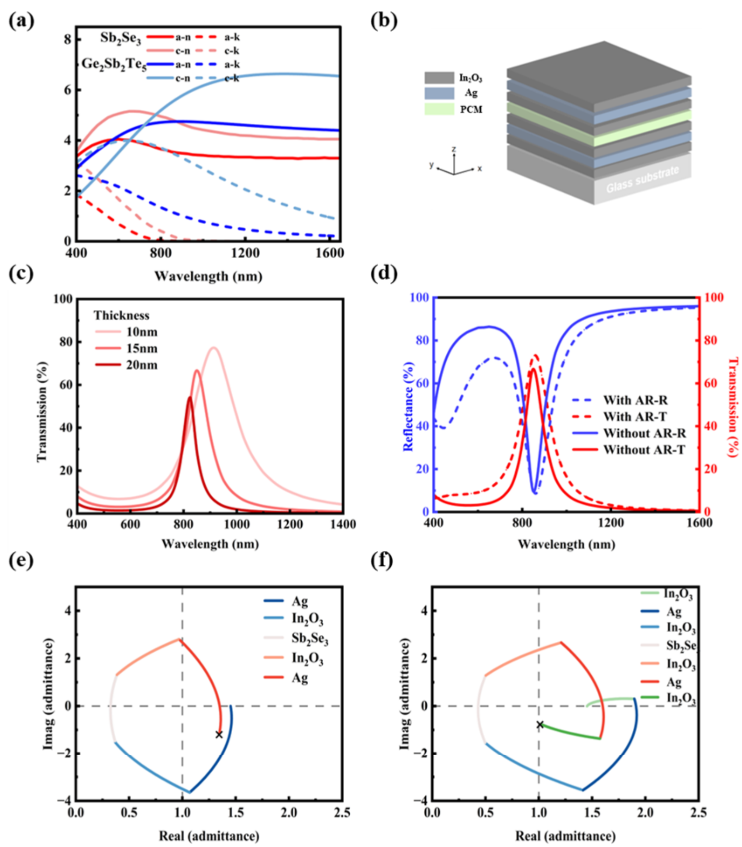

Fig. 1. (a) The optical constants of $Ge_2Sb_2Te_5$ and Sb_2Se_3 in the wavelength range of 400–1650 nm. (b) Three-dimensional schematic diagram of the multilayer film structure. (c) The transmission spectrum obtained through simulation when the thickness of the Ag layer is 10 to 20 nm. (d) Comparison of the transmittance and reflectance spectra with and without the In_2O_3 buffer layers. The optical admittance diagrams of the entire structure (e) with In_2O_3 buffer layers and (f) without In_2O_3 buffer layers.

Table 1. The thickness of the multilayer film structure

Material	In_2O_3	Ag	In_2O_3	Sb_2Se_3	In_2O_3	Ag	In_2O_3
Thickness/nm	30	15	30	10~300	30	15	70

Firstly, the thickness of the Sb_2Se_3 layer is fixed at 30 nm. The influence of the thickness of the Ag layer acting as the semi-transparent and semi-reflective mirror, as well as the In_2O_3 buffer layers, on the transmission spectrum will be studied. As shown in Fig. 1(c), the NIR transmission spectra of the multilayer structure are presented when the thickness of the Ag film is 10 nm, 15 nm,

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and 20 nm respectively. It can be observed that due to the large intrinsic absorption caused by the high carrier concentration of Ag in the NIR band, the peak transmittance significantly decreases as the thickness of the film increases. However, at the same time, thicker Ag layer enhances the reflectivity of the film, significantly increasing the resonance intensity. The half-width of the transmission peak is greatly reduced, and the transmission peak becomes sharper. As shown in Fig. 1(d), it demonstrates the influence of the 30 nm top In_2O_3 buffer layer and the 70 nm bottom In_2O_3 buffer layer on the NIR reflection and transmission spectra of the multilayer film. From the reflection spectrum, it can be seen that the In_2O_3 buffer layers significantly reduces the reflectivity of the multilayer film in the 450-800 nm range, while its anti-reflection effect is limited at 1000-1600 nm. This may be attributed to the fact that both Ag and In_2O_3 exhibit some intrinsic absorption in the NIR band. For the transmitted spectrum, the In_2O_3 buffer layers increased the peak transmittance from 66% to 73%, and the peak red shift was 3 nm. In order to further analyze the influence of the In_2O_3 buffer layers on the transmittance of the multilayer film, the optical admittance of the multilayer structure was calculated using the *Admittance Parameters* function in the *Tools* → *Analysis* module of the Essential Macleod simulation software, with SiO_2 selected as the substrate. As shown in Fig. 1(e),(f), the optical admittance of the transmission peak at 855 nm when the thickness of the Ag layer is 15 nm was calculated. For the optical admittance of the film, the closer the coordinate of the end point of the admittance trajectory is to (1,0), the more the equivalent admittance of the film approaches that of air, resulting in a lower reflectivity. As shown in Fig. 1(e), the admittance trajectory of the film starts at (1.45, 0) and ends at (1.35, -1.12), which is far away from the (1, 0) point of the air admittance. As shown in Fig. 1(f), after adding the In_2O_3 buffer layers, the end point of the admittance trajectory moves from (1.35, -1.12) to (1.01, -0.75), which is already very close to the air admittance. The 30 nm top layer and 70 nm bottom layer In_2O_3 films can effectively mitigate the refractive index difference between the exposed Ag layer, air, and SiO_2 substrate, making the optical admittance of the multilayer film closer to that of the air, thereby enhancing the peak transmittance.

As shown in Fig. 1(a), the optical constants indicate that Sb_2Se_3 exhibits extremely low absorption in the NIR wavelength, so it can be directly used as a NIR transparent material to construct the F-P cavity structure. This provides greater design freedom for the optical functional film of Sb_2Se_3. For the F-P cavity structure, when the optical path difference within the cavity is an integer multiple of a certain wavelength, the light in this wavelength will undergo constructive interference. Therefore, when the thickness of the optical cavity is large enough, the increase in the optical path difference will result in multi-order interference [45]. Thereby generating multiple transmission peaks. Optical films that have multiple transmission peaks in the NIR band can separate or combine light signals from two different wavelength bands, and process the information from both bands simultaneously. Combining the peak position modulation capability of the Sb_2Se_3 optical cavity, this tunable NIR filter has extremely high application prospects in fields such as gas sensors, laser beam combining, and multi- spectral imaging.

Therefore, we adjust the thickness of the Sb_2Se_3 layer from 10 nm to 300 nm. To study the transmission and peak position modulation capabilities of this multilayer film when it has multiple resonant wavelengths. As shown in Fig. 2(a), the visible and NIR transmission spectra of multilayer films with Sb_2Se_3 thicknesses of 40 nm, 160 nm, and 260 nm were simulated. Thermal images of the transmission distribution for the crystalline and amorphous states with Sb_2Se_3 thicknesses ranging from 40 to 300 nm are shown in Fig. 2(b). It can be observed that when the thickness of Sb_2Se_3 is 40 nm, the first transmission peak (1st) appears at 941 nm, with a transmission rate as high as 71%. When the thickness of Sb_2Se_3 is 160 nm, the position of the first peak shows a significant red shift, and the second peak appears at 890 nm, with a transmittance as high as 66%. And, even though Sb_2Se_3 has a thickness of up to 160 nm, it still has a 50% transmittance in crystalline state. This is attributed to its low NIR absorption in the crystalline state. Among them, the first peak exhibits the peak modulation of over 400 nm after the phase transition. The transmission peak shifts from 1833nm to 2236 nm. When the thickness of Sb_2Se_3 continued to increase to 260 nm, the first peak shifted beyond the wavelength of 2500 nm, and the third peak appeared at 831 nm. Meanwhile, the peak position of the second peak changed from 1227 nm to 1484 nm after the phase transition, shifting by 257 nm. Referring to Fig. 2(b), it can be observed that when the thickness of Sb_2Se_3 exceeds 100 nm and 200 nm respectively, the second and third transmission peaks gradually appear. At the same time, the red shift of the transmission peak and the modulation range of the peak position have increased. This is due to the increase in the optical path difference resulting from the increase in the thickness of Sb_2Se_3.The above results only changed the thickness of Sb_2Se_3. Therefore, we believe that after specifically designing and optimizing the thicknesses of Ag and In_2O_3, better results will be achieved in the corresponding wavelength range. Therefore, this multilayer film structure not only exhibits a high transmittance, but also enables customized design of multi-channel filter, as well as dynamic modulation of the peak positions.

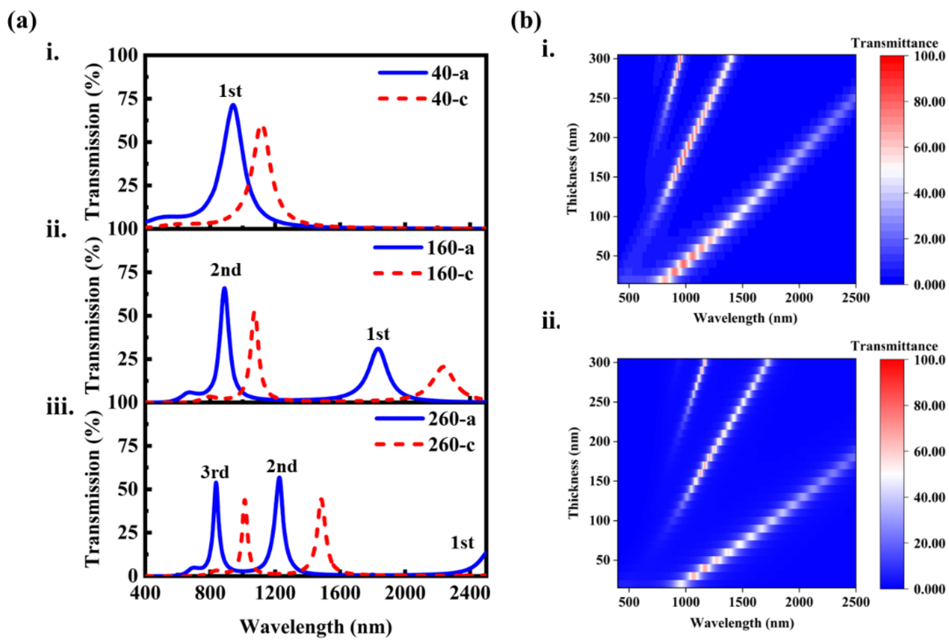

Fig. 2. (a) The transmission spectra were obtained through simulation for the PCM layer thicknesses of (i) 40 nm, (ii) 160 nm, and (iii) 260 nm. (b) When the PCM thickness is 10 to 300 nm the transmission spectra of (i) the amorphous state and (ii) the crystalline state.

In order to further analyze the generation mechanism of the double transmission peaks, the electric field distributions of multilayer films with thicknesses of 40 nm and 260 nm (having higher transmittance) were simulated. As shown in Fig. 3, the distribution wavelengths of the strong electric field in both amorphous and crystalline states are highly consistent with the position of the transmission peak, and they are mainly distributed in the Sb_2Se_3 layer. It is worth noting that when the thickness of Sb_2Se_3 is 40 nm, the electric field distribution of the Sb_2Se_3 layer within a single area. When the thickness of Sb_2Se_3 is 260 nm, the electric field at 831 nm shows a distribution pattern of strong - weak - strong, while the electric field at 1227 nm is distributed within a single area. This proves the existence of multi-order interference. As a result, the short wavelength (831 nm) has more than one resonant position, manifesting as a strong-weak-strong distribution of electric fields. Therefore, the above results indicate the influence of increasing the thickness of Sb_2Se_3 on the resonant mode of the film, which is consistent with the resonance mechanism of the F-P cavity.

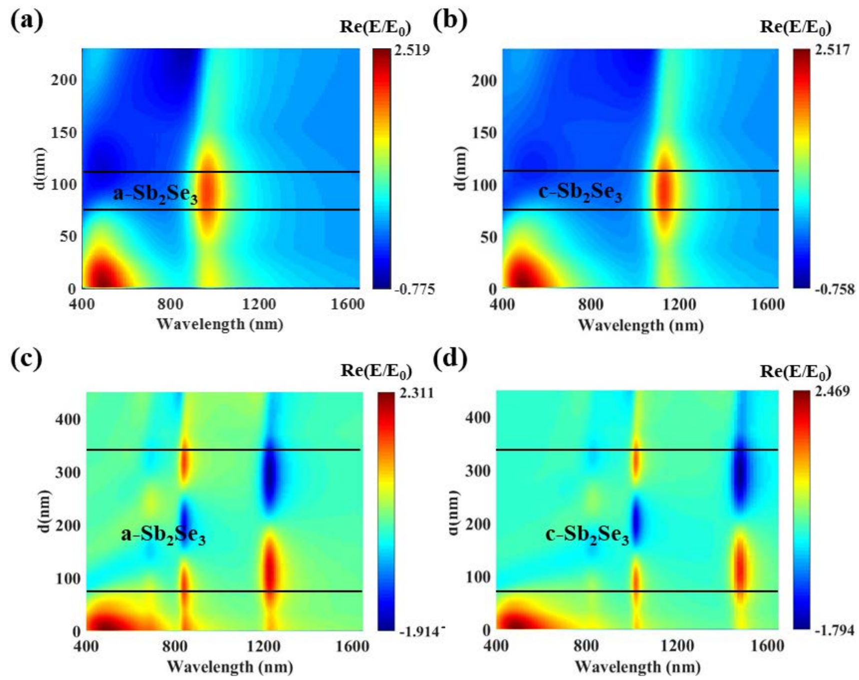

Fig. 3. The electric field distribution of amorphous (left) and crystalline (right) states of the multilayer film at Sb_2Se_3 thicknesses of (a),(b) 40 nm and (c),(d) 260 nm.

Furthermore, this multilayer film structure also exhibits excellent angle insensitivity. As shown in Fig. 4(a)-(d), the simulated transmission spectra of the multilayer film in the amorphous and crystalline states under different incident angles, when the thickness of Sb_2Se_3 is 30 nm and 260 nm, respectively. It can be observed that as the incident angle gradually increases from 0 to 80 degrees, the position of the transmission peak shifts towards the blue, and the peak transmission rate gradually decreases. As shown in Fig. 4(a), when the incident angle reaches 40°, the transmitted spectrum can still maintain the high transmission rate. The transmission rate decreases by 2% and the peak position shifts by 20 nm. When the incident angle exceeds 60°, significant distortion appears in the transmitted spectrum. This excellent angle insensitive behavior originates from the high refractive index of Sb_2Se_3 [46]. As shown in Fig. 4(c),(d), when the thickness of Sb_2Se_3 increases to 260 nm, similar angle insensitive behavior is maintained within an incident angle range of up to 40°. The main difference is that, at an incident angle of 80°, the transmission peak at longer wavelengths exhibits a larger spectral shift. This behavior is likely due to the increased dielectric thickness, which leads to larger variations in the optical path difference as the incident angle changes, thereby amplifying the resonance wavelength shift.

As shown in Fig. 5(a), this multilayer structure was fabricated using magnetron sputtering, and the measured NIR spectra of the crystalline and amorphous states of the Sb_2Se_3 layer at thicknesses of 25 nm, 35 nm, and 55 nm were obtained. Among them, the amorphous state represents the film deposition state, while the crystalline state is obtained by annealing at 310°C in nitrogen atmosphere for 30 min. It can be observed that when the thickness of the Sb_2Se_3 layer is 35 nm, the multilayer film has a peak transmittance of 55% in the amorphous state, a peak transmittance of 48% in the crystalline state, and the transmission peak position shifts from 921 nm to 1089 nm. This characteristic of no significant decrease in transmittance and

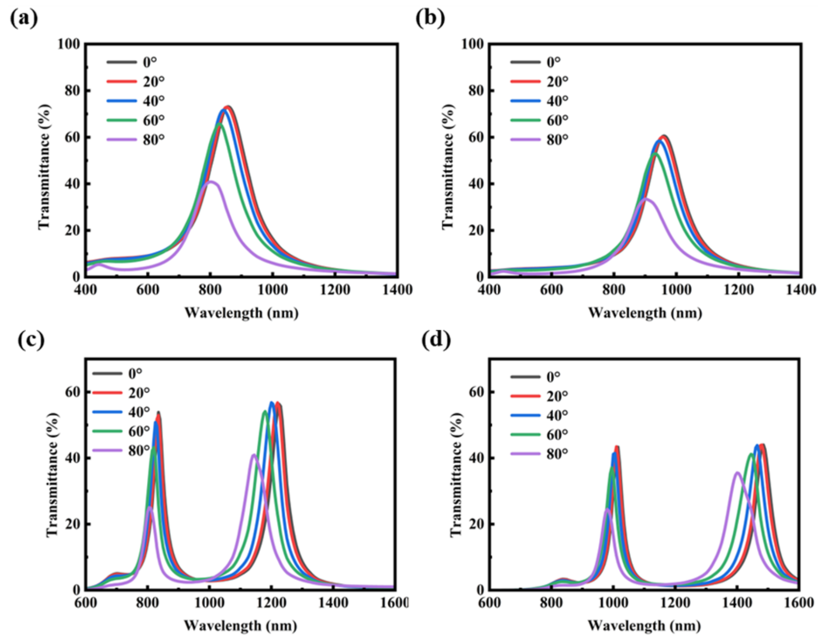

Fig. 4. The simulation transmission spectra of the multilayer film under different incident angles for Sb_2Se_3 thicknesses of 30 nm and 260 nm: (a) amorphous and (b) crystalline states at 30 nm; (c) amorphous and (d) crystalline states at 260 nm.

modulation range of the peak position exceeding 150 nm after the phase transition is rarely observable in traditional $Ge_2Sb_2Te_5$ materials. Furthermore, the position of the transmission peak of this multilayer film structure is significantly dependent on the thickness of the Sb_2Se_3, which is in excellent agreement with the simulation results. When the thickness of the Sb_2Se_3 layer increases, the transmission peaks of both the amorphous and crystalline states shift towards the longer wavelength, while maintaining the peak position modulation capability of over 150 nm. And when the thickness increases, the decrease in peak transmittance is likely related to the relatively large intrinsic absorption of the In_2O_3 thin film prepared by magnetron sputtering. In addition, as shown in Table 2, we have calculated the key filter characteristics, including the FWHM, full insertion loss across the band, and out-of-band rejection, providing a quantitative evaluation of the filter performance.

Table 2. The FWHM, full insertion loss across the band, and out-of-band rejection of multilayer film structure

Samples	FWHM (nm)	Insertion loss (dB)	Out-of-band rejection (dB)
25 nm-a	138	2.42	6.85
25 nm-c	82	3.28	8.27
35 nm-a	106	2.64	8.91
35 nm-c	104	3.23	12.14
55 nm-a	134	3.05	7.81
55 nm-c	146	4.61	10.76

As shown in Fig. 5(b), it presents the refractive index and extinction coefficient of the In_2O_3 film measured by the ellipsometer. In the range of 1000-2000 nm, the extinction coefficient of In_2O_3 significantly increases. This might be due to the fact that the large number of internal

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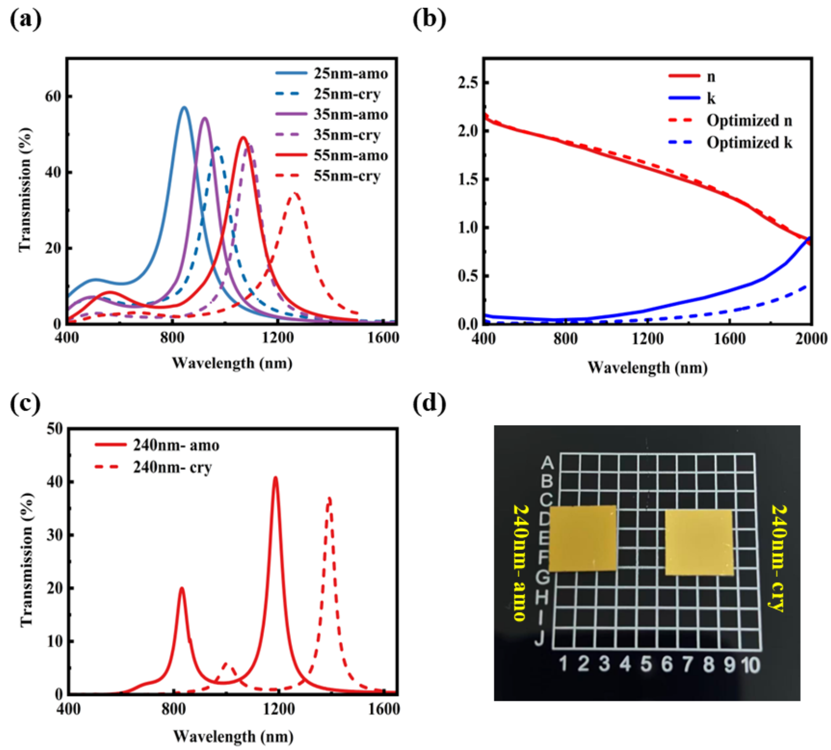

Fig. 5. (a) The measured visible and NIR transmission spectra when the thickness of Sb_2Se_3 is 25 nm, 35 nm, and 55 nm. (b) The optical constants of In_2O_3 film grown by magnetron sputtering. (c) The measured visible and NIR transmission spectra when the thickness of Sb_2Se_3 is 240 nm, and (d) the photograph of the fabricated sample.

defects were generated during the growth process of the film. This is because the optical loss of sputtered In_2O_3 films is highly sensitive to deposition parameters, where higher sputtering power and pressure can introduce defect and vacancies, leading to enhanced intrinsic absorption in the NIR region [47]. In addition, DC sputtering on insulating SiO_2 substrates may suffer from charge accumulation and intermittent arcing, which can further degrade the optical quality of the films [48].

To mitigate these effects, pulsed sputtering was employed and the sputtering pressure was reduced from 0.5 Pa to 0.3 Pa. As shown in Fig. 5(b), the extinction coefficient of In_2O_3 in the 1000–2000nm range is significantly reduced under the optimized conditions. On this basis, a filter with an Sb_2Se_3 layer thickness of 240 nm was fabricated to experimentally verify the multi-channel filtering capability. As shown in Fig. 5(c), two transmission peaks are observed at 830 nm and 1185 nm with peak transmittance exceeding 40%. After the phase transition, these transmission peaks redshift to 1002 nm and 1392 nm, respectively. Figure 5(d) presents photograph of the sample in the as-deposited and annealed states. No film cracking or delamination is observed after annealing, despite the 7-layer structure and the relatively thick Sb_2Se_3 layer.

4. Conclusion

In conclusion, we have successfully designed and fabricated the In_2O_3-Ag-In_2O_3-Sb_2Se_3-In_2O_3-Ag-In_2O_3 film stack, demonstrating the exceptional potential of Sb_2Se_3 as a low-loss, tunable medium material for high-performance NIR F-P cavities. The fabricated sample achieves

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the remarkable peak transmittance of up to 57% and exhibits the resonance wavelength shift exceeding 200 nm upon the phase transition of Sb_2Se_3. Furthermore, by simply increasing the Sb_2Se_3 layer thickness, this structure can also generate multiple resonant peaks, thereby achieving the function of a multi-channel narrow-band filter. Sb_2Se_3 exhibits excellent performance and multi-dimensional design flexibility in the NIR wavelength range, which contrasts sharply with the high loss and narrow tuning range of traditional $Ge_2Sb_2Te_5$, making it an ideal material for dynamic NIR photonics. This tunable multi-channel spectrally selective filter presents significant application potential in advanced fields such as multi-gas sensing, laser beam combining, and adaptive multi-spectral imaging.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. C. Fabry and A. Pérot, "Sur les franges des lames minces argentées et leur application à la mesure de petites épaisseurs d'air," Ann. Chim. Phys **12**, 459–501 (1897).
2. D. Wang, Z. Liu, H. Wang, *et al.*, "Structural color generation: from layered thin films to optical metasurfaces," Nanophotonics **12**(6), 1019 (2023).
3. C. Zhu, H. Zheng, L. Ma, *et al.*, "Advances in Fiber-Optic Extrinsic Fabry–Perot Interferometric Physical and Mechanical Sensors: A Review," IEEE Sens. J. **23**(7), 6406 (2023).
4. Z. Wang, X. Wang, S. Cong, *et al.*, "Towards full-colour tunability of inorganic electrochromic devices using ultracompact fabry-perot nanocavities," Nat. Commun. **11**(1), 302 (2020).
5. R. S. Daqiqeh, Z. Dong, E. C. J. You, *et al.*, "Nanophotonic Structural Colors," ACS Photonics **8**(1), 18–33 (2021).
6. Z. Yu, K. Zhao, Y. Zhao, *et al.*, "Hybrid SPR/Fabry–Pérot Cavity-Based Films with Versatile Color Tunability and Sensitive Environmental Sensing," Nano Lett. **25**(26), 10634 (2025).
7. J. Wng, X. D. Guo, C. J. Li, *et al.*, "Tunable MXene/WO_3 Fabry-Perot Microcavity Architecture for Bioinspired Soft Electrochromic Actuators with Vivid Colors," Adv. Funct. Mater. **35**(10), 2416812 (2025).
8. Z. M. Corato, M. A. Gil, X. Li, *et al.*, "Widely tunable and narrow-linewidth chip-scale lasers from near-ultraviolet to near-infrared wavelengths," Nat. Photonics **17**(2), 157 (2023).
9. L. J. Niu, X. Guo, X. Li, *et al.*, "Optofluidic Fabry-Perot Laser With Plasmon-Enhanced Tunable Emission," J. Lightwave Technol. **43**(15), 7193–7201 (2025).
10. Z. Zhang, F. L. Song, K. X. Xu, *et al.*, "Single-Mode Surface-Emitting Polariton Lasing with Switchable Polarization in a $CsPbBr_3$ Microwire Folded Fabry-Perot Cavity," Acs Photonics **11**(3), 1085–1092 (2024).
11. Z. Lang, S. Qiao, Y. He, *et al.*, "Disturbance-immune, fast response LITES gas sensor based on out-plane vibration mode employing micro Fabry-Perot cavity with heterodyne phase demodulation," Sens. Actuators, B **419**, 136412 (2024).
12. Z. Zhang, H. Gong, C. Yu, *et al.*, "An optical fiber humidity sensor based on femtosecond laser micromachining Fabry-Perot cavity with composite film," Opt. Laser Technol. **150**, 107949 (2022).
13. S. Abdollahramezani, O. Hemmatyar, H. Taghinejad, *et al.*, "Tunable nanophotonics enabled by chalcogenide phase-change materials," Nanophotonics **9**(5), 1189–1241 (2020).
14. P. Hosseini, C. D. Wright, and H. Bhaskaran, "An optoelectronic framework enabled by low-dimensional phase-change films," Nature **511**(7508), 206–211 (2014).
15. R. Z. Zhao, K. Guo, R. J. Zhu, *et al.*, "650 ps SET speed in $Ge_2Sb_2Te_5$ phase change memory induced by TiO_2 dielectric crystal plane," Infomat **6**(9), e12598 (2024).
16. J. B. Shen, W. X. Song, K. Ren, *et al.*, "Toward the Speed Limit of Phase-Change Memor," Adv. Mater. (Weinheim, Ger.) **35**(11), 2208065 (2023).
17. M. L. Martin, C. C. Popescu, L. Ranno, *et al.*, "Endurance of chalcogenide optical phase change materials: a review," Opt. Mater. Express **12**(6), 2145 (2022).
18. J. Meng, Y. Gui, B. M. Nouri, *et al.*, "Electrical programmable multilevel nonvolatile photonic random-access memory," Light:Sci. Appl. **12**(1), 189 (2023).
19. C. Q. Hu, L. Ma, X. C. Li, *et al.*, "Full-color, multi-level transmittance modulators: From reflectivity/gradient absorption coupling mechanism to materials map," Acta Mater. **216**(1), 117132 (2021).
20. Y. Zhao, H. Ji, M. Lu, *et al.*, "Thermochromic Smart Windows Assisted by Photothermal Nanomaterials," Nanomaterials **12**(21), 3865 (2022).
21. Z. Z. Duan, W. J. Chen, Z. X. He, *et al.*, "Reflective Full-Color Structural Colors with High Brightness and High Purity Based on $Ge_2Sb_2Se_4Te_1$ Chalcogenide Film," Adv. Opt. Mater. **12**(4), 2301577 (2024).

22. O. Hemmatyar, S. Abdollahramezani, T. Brown, *et al.*, "Phase-Change Material Micro-Displays," in *Conference on Lasers and Electro-Optics* (2021).
23. P. Prabhathan, K. V. Sreekanth, J. H. Teng, *et al.*, "Electrically Tunable Steganographic Nano-Optical Coatings," Nano Lett. **23**(11), 5236–5241 (2023).
24. Y. S. Huang, C. Y. Lee, M. Rath, *et al.*, "Tunable structural transmissive color in fano-resonant optical coatings employing phase-change materials," Mater. Today Adv. **18**, 100364 (2023).
25. D. Bombara, C. Williams, S. Bogr, *et al.*, "Automated real-time spectral characterization of phase-change tunable filters using a linear variable filter and IR camera for wide-field MWIR imaging," Opt. Eng. **60**(08), 084105 (2021).
26. H. Peelaers, E. Kioupakis, and D. W. C. G. Van, "Fundamental limits on optical transparency of transparent conducting oxides: Free-carrier absorption in SnO_2," Appl. Phys. Lett. **100**(1), 011914 (2012).
27. C. Q. Hu, Z. J. Zhou, X. Y. Zhang, *et al.*, "Far-infrared transparent conductors," Light:Sci. Appl. **12**(1), 98 (2023).
28. Z. R. Fang, J. J. Zheng, A. Saxena, *et al.*, "Non-Volatile Reconfigurable Integrated Photonics Enabled by Broadband Low-Loss Phase Change Material," Adv. Opt. Mater. **9**(9), 2002049 (2021).
29. S. T. Ilie, J. Faneca, I. Zeimpekis, *et al.*, "Thermo-optic tuning of silicon nitride microring resonators with low loss non-volatile Sb_2S_3 phase change material," Sci. Rep. **12**(1), 17815 (2022).
30. K. Murali, L. Thekkekara, M. A. Rahman, *et al.*, "Tunability of Sb_2Se_3 phase change material for multi-domain optoelectronics," Appl. Mater. Today **40**, 102338 (2024).
31. S. X. Gao, H. N. Ren, J. Z. Pang, *et al.*, "Reconfigurable Free Form Silicon Photonics by Phase Change Waveguides," Adv. Opt. Mater. **13**(15), 2402997 (2025).
32. C. M. Song, Y. X. Gao, G. X. Wang, *et al.*, "Compact nonvolatile 2×2 photonic switch based on two-mode interference," Opt. Express **30**(17), 30430 (2022).
33. M. Y. Wang, J. S. Lee, S. Aggarwal, *et al.*, "Varifocal Metalens Using Tunable and Ultralow-loss Dielectrics," Adv. Sci. (Weinheim, Ger.) **10**(6), 2204899 (2023).
34. M. Delaney, I. Zeimpekis, D. Lawson, *et al.*, "A New Family of Ultralow Loss Reversible Phase-Change Materials for Photonic Integrated Circuits: Sb_2S_3 and Sb_2Se_3," Adv. Funct. Mater. **30**(36), 2002447 (2020).
35. C. Rios, Q. Y. Du, Y. F. Zhang, *et al.*, "Ultra-compact nonvolatile phase shifter based on electrically reprogrammable transparent phase change materials," Photonix **3**(1), 26 (2022).
36. M. Delaney, I. Zeimpekis, H. Du, *et al.*, "Nonvolatile programmable silicon photonics using an ultralow-loss Sb_2Se_3 phase change material," Sci. Adv. **7**(25), eabg3500 (2021).
37. J. Faneca, I. Zeimpekis, S. T. Ilie, *et al.*, "Towards low loss non-volatile phase change materials in mid index waveguides," Neuromorph. Comput. Eng. **1**(1), 014004 (2021).
38. Z. R. Fang, C. Rui, J. J. Zheng, *et al.*, "Ultra-low-energy programmable non-volatile silicon photonics based on phase-change materials with graphene heaters," Nat. Nanotechnol. **17**(8), 842–848 (2022).
39. G. D. A. Ramos, A. E. Gallegos, R. C. A. Salcedo, *et al.*, "Design of a tunable Fabry-Perot filter based on a silicon wafer for gas sensing applications in the infrared," Infrared Physics & Technology **145**, 105689 (2025).
40. Y. Kim, S. G. Goo, and J. S. Lim, "Multi-Gas Analyzer Based on Tunable Filter Non-Dispersive Infrared Sensor: Application to the Monitoring of Eco-Friendly Gas Insulated Switchgears," Sensors **22**(22), 8662 (2022).
41. X. H. Xu, G. S. Li, B. Xie, *et al.*, "Multi-channel laser beam combining and closed-loop correction technology in visible light band," Chinese Optics **17**(2), 342–353 (2024).
42. Z. Ye, H. Miao, X. Liu, *et al.*, "Experimental Study on High- Power Laser Beam Combining Using Dichroic Mirrors," Laser Optoelectron. Prog. **60**(15), 1514002 (2023).
43. Y. Q. Zhu, X. Tian, J. K. Fang, *et al.*, "Independently tunable all-dielectric synthetic multi-spectral metamaterials based on Mie resonance," RSC Adv. **12**(32), 20765–20770 (2022).
44. D. Silva, H. Kala, M. Z. D. Tripathi, *et al.*, "Technologies for adaptive SWIR multispectral imaging; proceedings of the 2024 Conference on Electro-Optical and Infrared Systems," Electro-Optical and Infrared Systems: Technology and Applications **13200**, 41 (2024).
45. C. Wang, J. Wang, J. Wang, *et al.*, "Research on optimal design of ultra-bandwidth multi-channel filter based on Fabry-Perot structure," Results Op **4**, 100086 (2021).
46. C. Y. Yang, W. D. Shen, Y. G. Zhang, *et al.*, "Compact Multilayer Film Structure for Angle Insensitive Color Filtering," Sci. Rep. **5**(1), 9285 (2015).
47. D. S. Ginley and J. D. Perkins, "Transparent Conductors," *Handbook of Transparent Conductors* (Springer, 2011), 1–25.
48. G. Rane, P. Durrenfeld, U. Krause, *et al.*, "Enhanced control of cathode arcing during sputter deposition of insulating films," Surf. Coat. Technol. **512**, 132409 (2025).